

DDR基础课程

第九节 - DDR PHY基本结构

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目录

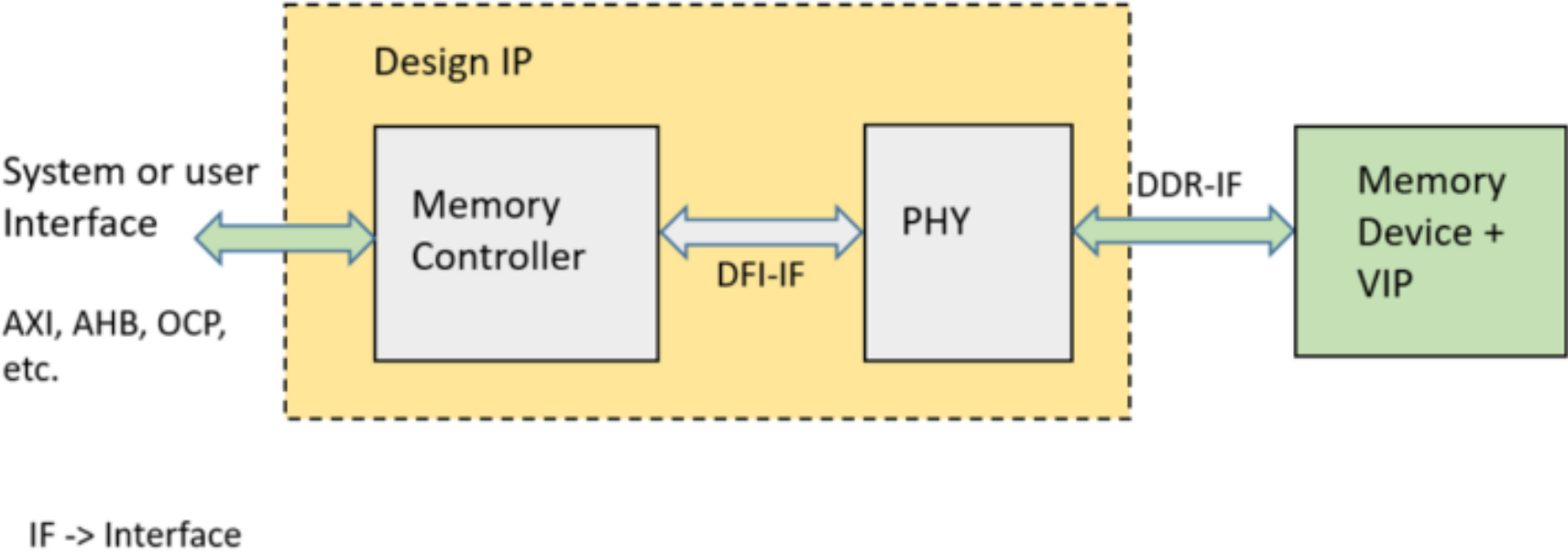
关注的问题

- DFI接口基本介绍
- DDR PHY的主要功能目标
- 典型常见特性
- 主要架构划分

DFI接口基础介绍

- DFI - DDR PHY Interface
- DDR Controller 与 DDR PHY之间的接口

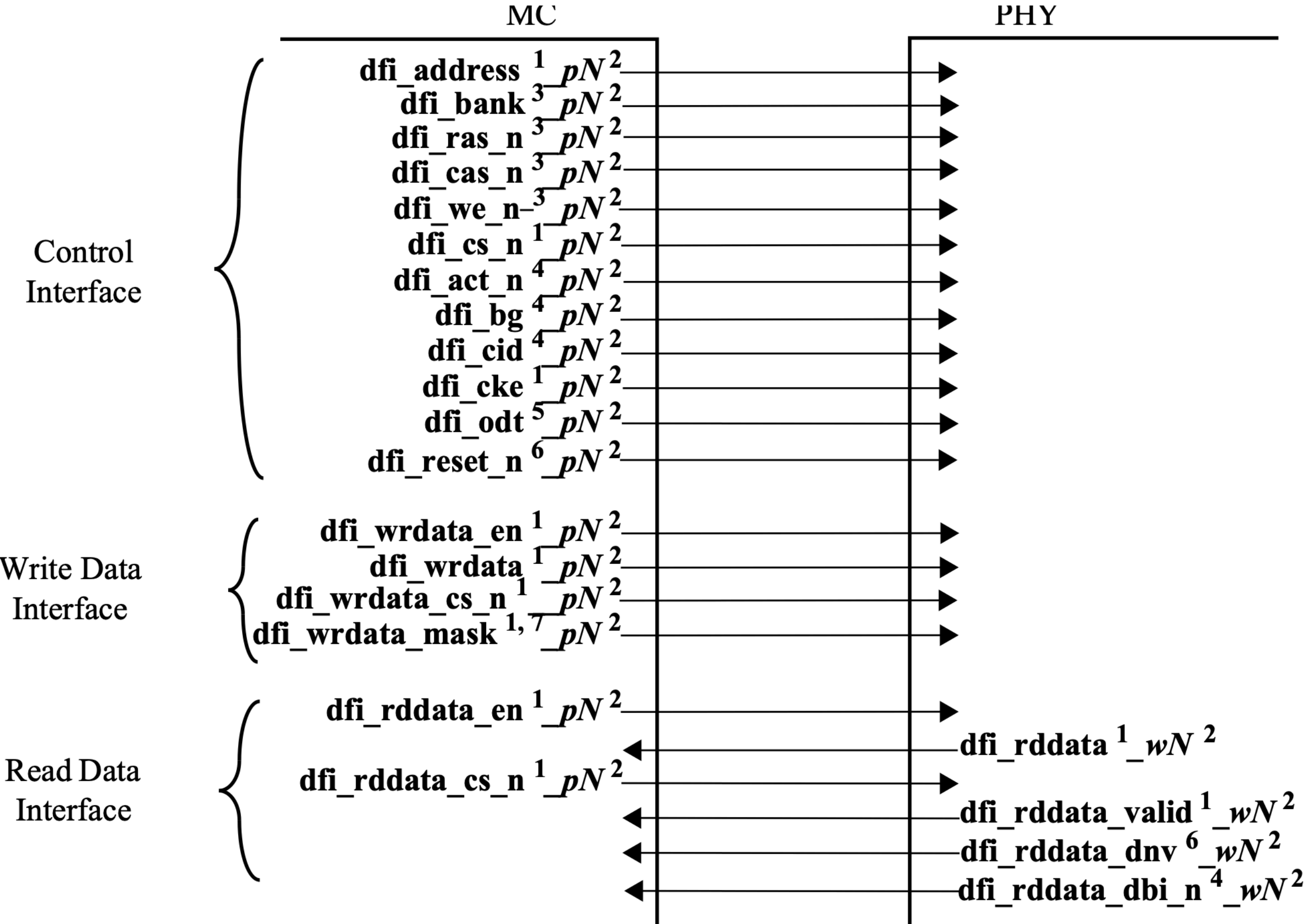
Memory subsystem



Interface Group	Description
Control	Required to drive the address, command, and control signals to the DRAM devices.
Write Data	Used to send write and receive valid read data across the DFI.
Read Data	
Update	Provides an ability for the MC or the PHY to interrupt and idle the DFI.
Status	Used for system initialization, feature support, and to control the presence of valid clocks to the DRAM interface.
Training	Used to execute gate training, read data eye training, and write leveling operations.
Low Power Control	Allows the PHY to enter power-saving modes.
Error	Used to communicate error information from the PHY to the MC.

DFI接口基础介绍

- phase用于处理低频与高频之间的转换问题
- 低频 - DFI Clock域
- 高频 - DRAM CK/WCK域
- Example
 - LPDDR5 6400Mbps
 - DRAM WCK 3200MHz
 - PHY High Speed Clock 3200MHz
 - Memory Controller 800MHz
 - Clock Ratio = 1:4
 - Phase number = 4



DDR PHY的主要功能目标

- 接口转换，即DFI接口与DDR接口的转换
- 频域转换，即DFI clock domain与DDR clock domain的转换
- SI - Signal Integrity
- 初始化 & Training
- ReTrain

典型常见特性

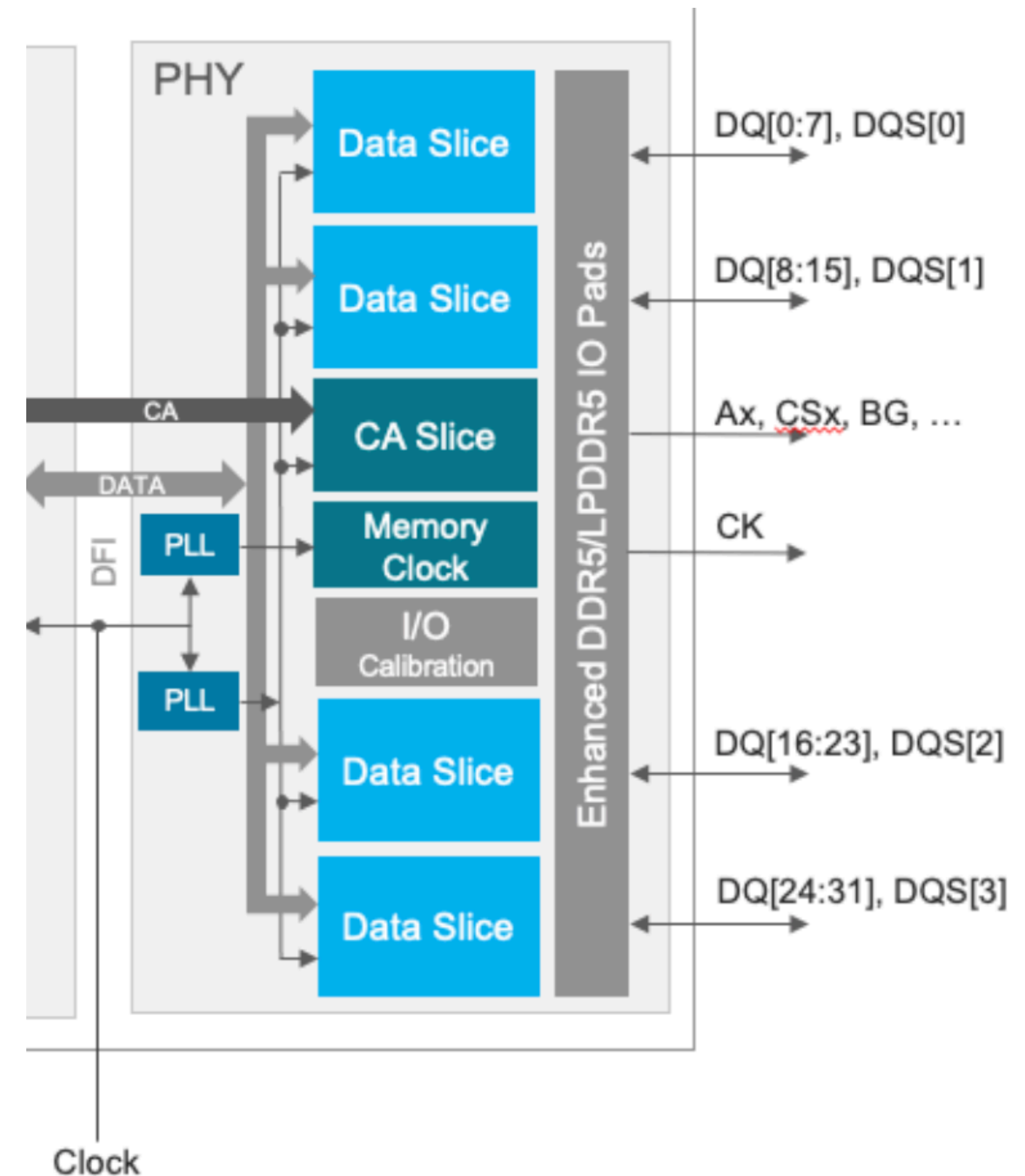
- Rambus
- Synopsys

Features	Deliverables	Services
• PLL-based clocking with internal clock alignment to the parallel clock on the memory controller interface • Autonomous initialization • Support for multiple channel configurations from x16 through x72 per channel • Support for multiple DRAM widths (x4, x8, x16, x32) • Support for single channel, 1 to 4 ranks • Excepteur sint occaecat cupidatat non proident, • Selectable low-power operating states • DFI 3.1 compliant for easy integration with memory controller • Programmable output impedance and on-die termination • ZQ calibration of output impedance and on-die calibration • Utilizes standard 8-layer 6020 metal layer stack. • Supports C4 flip-chip packaging options • Register interface for state observation • Test traffic generation and error checking for in-situ test • LabStation™ software environment for system level bring-up, characterization, and validation		

Highlights	Products	Downloads and Documentation
▪ Supports JEDEC standard DDR5 and DDR4 SDRAMs ▪ High-performance DDR PHY supporting data rates up to 8400 Mbps ▪ PHY independent, firmware-based training using an embedded calibration processor ▪ Supports up to 4 trained states/ frequencies with <3µs switching time ▪ I/O receiver decision feedback equalization ▪ VT compensated delay lines for DQS centering, read/write 1D (DDR4) and 2D training (DDR5), and per-bit deskew on both read and write data paths ▪ DFI 5.0-compliant controller interface ▪ Designed for rapid integration with Synopsys memory controller for a complete DDR interface solution		

主要架构划分

- Cadence
- Digital PHY & Analog PHY
- PLL - Phase Locked Loop
- Data Slice
 - Parallel In Serial Out & Serial In Parallel Out
 - Async FIFO 用于跨时钟域处理
 - Read Latch - RDQS采样RDQ
- Initialization, Training, ReTrain
 - Hardware支持
 - Software Solution
 - Hardware Solution



上节课后问题解答

- 课后问题1: DDR Controller的频率为800MHz, DDR PHY的数据速率为6400Mbps。这种情况下, wdata/rdata需要4个phase? **Controller Clock 800MHz - PHY Clock 3200MHz。**
- 课后问题2: 为了保证seamless的数据访问, 调度的策略应当是以下哪种? A. 跨BG; B. 相同BG内; **A。跨BG的tCCD是tCCDS。**
- 课后问题3: 为了保证调度的公平性, 某个一直没有被调度出去的命令, 即使需要打断其他连续seamless的访问, 也应当被调度出去吗? 为什么会出现这种情况? **是。为了公平性, 不能任由某一个命令无限时间待在controller中而没有被调度出去。这种情形也称为starvation。这种情形发生的一个场景是, 此命令以外的命令长时间连续出现row-hit, 基于效率的考量, 会优先将row-hit的命令调度出去。**
- 课后问题4: 为了保证调度的效率, 是否应当尽可能经常性地地进行读写切换? **否。读写切换本身是有代价的, 读写切换之间的时序参数是比较大的。如果在一个方向（读方向或者写方向）可以连续有row-hit的命令, 则为了效率应当减小读写切换的频率。**
- 课后问题5: 上游地址连续对于DDR的访问效率是否友好? **是。上游的地址越连续, 则越容易映射出来row-hit的命令。**
- 课后问题6: read data channel如果被反压, 是否会影响到ddr的访问效率? **是。read data channel如果出现反压, 可能会导致read data buffer满。而read data buffer如果满了, 则会反压住read命令的发送, 因为这时候如果发送命令, 是没有地方来存放返回的读数据的。**
- 课后问题7: open page policy是否优于closed page policy? **否。哪种page policy好需要根据具体的场景来确定。**

课后问题

- 课后问题1: DFI接口用于连接___与___?
- 课后问题2: DFI接口上的phase用于解决_____的问题?
- 课后问题3: PHY的用途包括_____?
- 课后问题4: PHY做CA & DQ Training的目的是_____?
- 课后问题5: PHY做ReTrain的目的是_____?
- 课后问题6: PHY对于ZQ Calibration的需求是_____?
- 课后问题7: PHY的Training有hardware和software两种控制方式, 哪种更好? 为什么?

下节预告

第十节 - DDR课程总结及答疑

- 课程总结
- 问题回顾
- 答疑

谢谢